SOFTOMATIC
TECHNOLOGY SERVICES

TRAINER GUIDE · GENAI + PROJECTS · TUE 27 OCT

GenAI Session 5
Teach reliable LLM calls

> One lesson for 84 different projects: stream the answer, retry only what can succeed, fall back to a second model, and put a price on every question.

S3
12:45 to 2:15
Group A + Group T

S4 (SAME SESSION AGAIN)
2:15 to 3:45
Group B + Group T

STUDENT NOTES
GenAI05 PDF
share before 12:45

THE CLASS IN FIVE MOVES

01 Recap Monday, the 429 hook

02 Board: retry tree + backoff

03 Live-code client, fallback, cost

04 Students build; break it on purpose

05 **14 tests, exit ticket, homework**

THE 90-MINUTE PLAN

Run sheet for today

Teach the same 90 minutes twice. Group T sits in both: in the 2:15 repeat they finish the fallback tests and help one neighbour from Group B.

TIME	WHAT HAPPENS	YOUR GOAL
0–5	Recap Monday in 3 cold-call questions; tell the 429 demo story	Everyone knows why today matters
5–30	Board: streaming timeline, retry decision tree, backoff windows, fallback rules, cost formula	Students can say what to retry and what not
30–55	Live-code with students typing along: streaming client, fallback, cost log, /chat, /usage	Same code on every laptop
55–75	Students finish, run with the fake provider, then break it on purpose (wrong URL, wrong key)	Everyone has seen a retry and an error event
75–85	ruff check . and pytest -q: 14 passed; first real-model call for students with a key	Green tests on every laptop
85–90	Exit ticket + homework GEN5 (docs/COST.md) + reminder: assessment Wednesday afternoon	Every student knows tonight's task

Six things every student must leave with

Stream
SSE events; same total time, faster first word.

Timeout
Every call; 30 s means longest silence.

Retry rules
429/5xx/timeouts yes; 4xx never; never after text.

Backoff + jitter
random(0, min(8, 0.5 × 2^attempt)).

Fallback
Only on LLMUnavailable, only before text.

Cost + /usage
tokens × price per million; totals per day.

THE ONE IDEA TO HOLD ALL CLASS

Every student's product is different, but the LLM call is the **same engine**: the same client, the same retries, the same cost log. Only the questions they send change. Teach the engine once; make each student test it with a question from their own `docs/QUESTIONS.md`.

30 MINUTES THE EVENING BEFORE

Prepare your repo and demo

You live-code from your own copy of the starter repo. Everything you type is in the student notes (sections 07–13 and the appendix), and it has been tested: **ruff passes and pytest shows 14 passed** with the fake provider.

Checklist

- [ ] Your repo has Monday's files (config, schemas, tokens, llm/base.py, llm/fake.py). Type today's files once at home and run `pytest -q` → 14 passed. Then `git stash` them so you can type them live
- [ ] A key for the provider you chose for the batch, in `.env` only. Check its free-tier limits page tonight; limits change
- [ ] Practise the two "break it" demos on the next pages; each takes under a minute
- [ ] Classroom Wi-Fi: the provider's API and pypi.org open; `pip install httpx==0.28.1` works
- [ ] The GenAI05 student notes PDF shared in the group before 12:45; tell students who missed Monday to type the appendix files first
- [ ] The project tracker open, to note who finishes with 14 tests passing

DO NOT

Do not let 40 students hit one shared key at the same time in class. That is today's hook happening for real. Students test with the **fake provider**; only students with their own key call a real model, and only after tests are green.

STUDENTS WHO MISSED MONDAY

Seat them next to a Group T student. They type appendix A1–A4 (about 15 minutes) while you draw on the board, then join the live coding at minute 30.

OPEN, THEN DRAW THE RULES

Recap, hook and the board

Cold-call names for the recap. Tell the story in one minute. Then draw; students copy the two drawings into their notebooks.

ASK THE CLASS

Q. Why does /chat call an interface and not a vendor SDK?
Expect: So we can switch provider, use the fake one in tests, or add a fallback, by configuration only.

Q. Where does the API key live, and where must it never be?
Expect: In .env (git-ignored); never in code or on GitHub.

Q. What does the fake provider give us?
Expect: Tests and CI with no key, no quota, no internet.

SAY THIS

"It is your project demo. The interviewer types a question. 30 seconds of spinner, then 500 Internal Server Error. The log says 429: your batchmate used the same key."
"Today you make the call survive that: stream, retry the right errors, fall back, and know the price."

DRAW 1: WHAT TO RETRY

Failed call → text already sent? **yes → stop, event: error.** No → which error?
400 · 401 · 403 · 404 → **never retry** (fix your config)
timeout · connection · 429 · 500 · 502 · 503 · 504 → **wait, retry** (max 3)
Still failing → **LLMUnavailable** → fallback model, if one is set

DRAW 2: BACKOFF WINDOWS (BASE 0.5 S, CAP 8 S)

attempt 0: 0–0.5 s · 1: 0–1 s · 2: 0–2 s · 3: 0–4 s · 4+: 0–8 s
Wait = a **random** point inside the window (jitter). 3 retries → at most 0.5 + 1 + 2 = **3.5 s** of waiting.

ASK THE CLASS

Q. 90 students get 429 at the same moment and all retry after exactly 1 s. What happens?
Expect: They collide again: 90 calls in the same instant. Jitter spreads them out.

Q. A 1,200-token prompt and a 300-token answer at $0.50 / $1.50 per million. Cost?
Expect: $0.0006 + $0.00045 = $0.00105 per question, about $1.05 per 1,000.

LIVE CODING: STUDENTS TYPE ALONG

Live-coding plan, step by step

Project your editor. Type slowly, say each line's job aloud. Students type the same code from the notes; nobody copy-pastes whole files.

1. **pip install httpx==0.28.1 and add it to requirements.txt**

```
pip install httpx==0.28.1
```

Say: "The app now makes HTTP calls, so httpx is a runtime dependency."

2. **openai_compat.py part 1: RETRYABLE_STATUS and backoff_delay (notes section 07)**

Say: "This is a whitelist. Anything not on it is never retried."

3. **openai_compat.py parts 2–4: constructor, request, status checks, reading the stream**

Say: "Point at three lines: the timeout, raise LLMError for 4xx, and the produced_any check."

4. **base.py: two StreamResult lines + LLMUnavailable; fallback.py; factory branch; three settings (section 09)**

Say: "The wrapper is itself a provider. The route does not change."

5. **cost.py with fallback_used; main.py with /chat and /usage; usage.py (sections 11–12)**

Say: "Every call writes one line. /usage just adds them up per day."

6. **conftest.py and tests/test_fallback_and_usage.py (section 13), plus the four reference test files**

```
ruff check .
pytest -q        # 14 passed
```

Say: "Green with no key, no internet, in under a second."

7. **Run it with the fake provider and stream one question**

```
uvicorn app.main:app --reload
curl -N -X POST localhost:8000/chat -H "Content-Type: application/json" \
  -d '{"messages":[{"role":"user","content":"What is pgvector?"}]}'
curl localhost:8000/usage
```

Say: "Watch the words arrive one by one, then event: done with tokens and cost."

IF TIME RUNS SHORT

Do steps 1–3 and 5 live; students type step 4 (fallback) and step 6 (tests) on their own from the notes in minutes 55–75. Everything they need is printed.

SHOW FAILURES BEFORE STUDENTS MEET THEM

Break it on purpose

Two demos, under a minute each, then students repeat them on their own laptops. We ran both against the tested code; the output below is what you will see.

Demo A · a provider that is down (no internet needed)

Point the app at a port where nothing is listening. Every attempt gets a connection error, so the client retries with backoff and then gives up.

.ENV, THEN RESTART UVICORN AND CALL /CHAT

```
LLM_PROVIDER=openai_compat
LLM_BASE_URL=http://127.0.0.1:9/v1
LLM_MODEL=x

# what the browser / curl receives after about 3 seconds:
event: error
data: {"error": "LLM call failed after 4 attempts: All connection attempts failed"}

event: done
data: {"usage": {"input_tokens": 0, "output_tokens": 0, "estimated": false}, "cost_usd": 0.0}
```

ASK THE CLASS

Q. Why about 3 seconds and not 0?
Expect: 4 attempts; between them the random backoff waits, at most 0.5 + 1 + 2 = 3.5 s.

Q. Why is the HTTP status still 200?
Expect: The stream had already started; the error travels inside the stream as event: error.

Demo B · now add a fallback

Add `LLM_FALLBACK_MODEL=y` and call again. The fallback (same dead URL here) is tried with 1 retry. Open `logs/cost.jsonl` : the last line says `"model": "y", "fallback_used": true`. That proves the switch happened. To see it **succeed**, run `pytest -q -k fallback`, or point `LLM_FALLBACK_BASE_URL` and key at a working provider.

OPTIONAL DEMO C · WRONG KEY

With your real provider URL and a made-up key, the provider should answer 401: one attempt, an immediate `event: error`, and **no fallback**. Ask: "Why don't we fall back here?" Expect: "It is a config bug; a fallback would hide it."

SAME ENGINE, THEIR PRODUCT

What to check on each laptop

Go row by row. Every laptop: 14 tests green, one streamed answer to **their own** question, one line in logs/cost.jsonl. Then one engine-specific check.

R REG

- [ ] Question from QUESTIONS.md about a rule or circular
- [ ] Notes that input tokens will grow when chunks arrive (Week 2)

E ENT

- [ ] Question names a client ("leave policy at company A")
- [ ] Knows /usage could later be split per client

F FIN

- [ ] Question about a number in an annual report
- [ ] Says fewer, better chunks is the first cost lever

C CODE

- [ ] Question about a function or file in the repo
- [ ] Trusts provider usage numbers, not the 4-character estimate

C CIT

- [ ] Tries one question in Hindi, Tamil or Telugu
- [ ] Compares estimated vs reported tokens if they have a key

Red flags for every engine

- [ ] A real API key inside a .py file or in a commit
- [ ] A try/except that retries every error
- [ ] Tests that need the internet to pass
- [ ] Copy-pasted code they cannot explain line by line

USE GROUP T

In the 2:15 repeat, Group T students who are green pair with Group B students who are stuck. Most problems are a missing `__init__.py`, a typo in a setting name, or httpx not installed.

EXPECT THESE, HAVE THE ANSWER READY

Common errors and fixes

PROBLEM	WHAT TO TELL THE STUDENT
ModuleNotFoundError: app.llm	Missing `app/llm/__init__.py`; create an empty file.
ModuleNotFoundError: httpx	`pip install httpx==0.28.1` inside the activated .venv; add it to requirements.txt.
test_health fails: environment is "local"	They kept the old conftest. Use the new one: it sets `environment="test"`.
Tests try to call a real provider	The conftest fixture forces `llm_provider="fake"`; make sure tests use the `client` fixture.
ImportError: cannot import name LLMUnavailable	They skipped the base.py addition (notes section 09).
401 from the provider	Key and base URL do not belong together, or the key has a space. Not retried, no fallback, by design.
404 from the provider	Wrong model ID or base URL path; copy the model ID from the provider's model list.
400 mentioning stream_options	That provider rejects the field: remove it for that provider; usage becomes estimated.
Answer arrives all at once	Use `curl -N`; a browser extension or proxy may buffer.
429 in class	Shared key. Switch that student back to the fake provider; this is the hook, live.

RULE FOR KEYS

If a student pushed a real key to GitHub, they must revoke it on the provider's site today and create a new one. Deleting the commit is not enough: the key is already public.

CLOSE THE LOOP

Checkpoint, exit ticket and homework

Pick two students to show their test run and one line of logs/cost.jsonl on the projector. Read the line aloud: model, tokens, cost, fallback_used.

Checkpoint: done means

- [ ] ruff check . passes
- [ ] pytest -q shows 14 passed
- [ ] One of THEIR questions streamed through /chat
- [ ] logs/cost.jsonl has a line; /usage shows today
- [ ] Code pushed to GitHub

ASK THE CLASS

Q. Exit 1. Name two errors you retry and two you never retry.
Expect: Retry: 429, 503 (or timeout, connection error). Never: 401, 404 (or 400, 403).

Q. Exit 2. When does the fallback answer?
Expect: Only on LLMUnavailable, and only before any text reached the user.

Q. Exit 3. 1,000 input + 500 output tokens at $0.50 / $1.50 per million?
Expect: $0.00125.

Homework GEN5 · docs/COST.md (due before Thu 29 Oct GenAI)

CHECK	FULL MARKS WHEN	MARKS
Code pushed, CI green	ruff and 14 tests pass on GitHub Actions	3
10 own questions asked	From their docs/QUESTIONS.md; real model, or fake and stated	2
Numbers from /usage	Average input and output tokens per question, from the log, not guessed	2
Cost per question and per 1,000	Arithmetic correct for the prices stated, with the pricing page link	2
One way to cut cost	Specific to their product (fewer chunks, caching, cheaper model for easy questions)	1

SAY THIS BEFORE THEY LEAVE

"Tonight: push today's code with 14 tests green, then ask 10 of your own questions and write docs/COST.md. Wednesday afternoon is the assessment, so finish it tonight. Next GenAI is Thursday: Docker and Docker Compose."

TWO MINUTES AT THE END OF EACH PART

Interview questions to cold-call

All five are in the student notes (section 16) with model answers. All are labelled **PATTERN-ONLY**: common in GenAI interviews, no company claimed. Listen for the key points; push for numbers.

#	QUESTION	LISTEN FOR	WHO TO ASK
Q1	How do you handle rate limits and failures from an LLM API?	Timeout; retry only 429/5xx/timeouts; capped backoff + jitter; never after streaming; fallback; rate limits; caching.	Ask someone who looked confident in the demo.
Q2	Why stream LLM responses, and how?	Same total time, lower time to first token; SSE (one-way, plain HTTP); watch buffering proxies.	Good for a quieter student: it is in the drawing.
Q3	Which HTTP errors should a client retry?	Transient: timeouts, resets, 429, 502/503/504, usually 500; respect Retry-After; never 400/401/403/404; cap retries.	Follow up: "What is a retry storm?"
Q4	How much does one question cost in your app?	Tokens × price per million; ≈ $0.001 for 1,200 + 300 tokens at $0.50/$1.50; levers: fewer chunks, cache, cheaper model.	Ask a FIN or REG student: their prompts are the biggest.
Q5	How do you get reliable JSON out of an LLM? (Monday)	Schema + structured mode + Pydantic validation + one retry with the error + fallback; never act on unvalidated output.	Revision: ask someone who missed Monday.

Weak answers to push back on

"Wrap it in try/except and retry." · "Streaming is faster." · "Retry everything 5 times." · "It's free, I use a free tier."

Follow-up with a number

"Base 0.5 s, cap 8 s, 3 retries: worst-case waiting?" → 3.5 s plus the attempts.
"1 million questions a month at $0.00105?" → about $1,050.

MAKE IT THEIRS

After a good answer, ask: "Say it again for your own product, in one sentence." That is the version they will use in a real interview.

KEEP THIS ON THE DESK

One-page trainer cheat card

Times

S3 12:45–2:15 · Group A + T

S4 2:15–3:45 · Group B + T (repeat)

Live coding at minute 30; break-it demos at 55

Retry rules

Yes: timeout, connection, 429, 500/502/503/504

Never: 400, 401, 403, 404

Never after text was streamed

Backoff

random(0, min(8, 0.5×2^{attempt}))

Windows 0.5 · 1 · 2 · 4 · 8 s

3 retries ≤ 3.5 s of waiting

Fallback

`LLM_FALLBACK_MODEL` (+ optional URL, key)

Only LLMUnavailable, only before text

Backup gets max_retries=1

Cost

(in × price_in + out × price_out) / 1,000,000

1,200 + 300 tokens at $0.50/$1.50 = $0.00105

Estimates (4 chars/token) never bill

Demos

Dead URL: `http://127.0.0.1:9/v1` → 4 attempts, ~3 s

+ fallback → log shows fallback_used: true

`pytest -q` → 14 passed

Homework GEN5

docs/COST.md from 10 own questions

Due before Thu 29 Oct GenAI

Marked out of 10 (page 9)

Hooks

"Some Rains Fall, People Look"

"Double the window, pick a random seat"

"Unavailable = try someone else. Error = fix yourself."

TRAINER CHECKLIST

Four habits for this class

Draw before you code
The retry tree on the board makes the code obvious.

Fake provider in class
No shared-key 429s; real models only after tests are green.

Break it on purpose
Students trust retries once they have seen one.

Their question, their cost
Every student streams one question from their own project.

> By 3:45 every student has a **streaming, retrying, costed** LLM call with 14 tests passing, and can explain which errors it retries and why.

TODAY

BEFORE 12:45	12:45 · 2:15	TONIGHT
Share the GenAI05 PDF	Draw, live-code, break it	Code pushed, docs/COST.md